

Sophia Zhang

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EDUCATION

Massachusetts Institute of Technology

Bachelor's in CS and Math

Cambridge, MA

Intended: Aug. 2025 – May 2028

Princeton International School of Mathematics and Science (PRISMS)

High school

Princeton, NJ

Sep. 2022 – June 2025

EXPERIENCE

Quantitative Research Intern

Citadel Securities

Summer 2027

Miami, FL

- Incoming Intern

AI Engineering Intern

Alpha

June – Aug. 2026

Austin, TX

- Intensive program for top MIT and Stanford students. Worked on app dev, model training, evals, and leveraging agents for max productiveness for education and learning science based projects.

Undergraduate Research Opportunities Program

Dina Katabi's Lab at MIT CSAIL

Sep. 2025 – Present

Cambridge, MA

- Working with multimodal data, mostly EEG and breathing

Trading and Technology Internship

Hudson River Trading

Jan. 2026

New York City, NY

- Worked on many projects in both software engineering and quant research, including trading Brazilian equities.

LLMs for Formal Theorem Proving

Microsoft Research Asia

June 2025 – Jan. 2026

Remote

- Helped verify math proofs in Lean and discussed findings to improve the system.
- ICML 2026: Euclean: Automated Geometry Problem Formalization with Unified Verification in Lean
- STAR-PólyaMath: Multi-Agent Reasoning under Persistent Meta-Strategic Supervision

PROJECTS

AI Math Olympiad Progress Prize 3 (Kaggle) | *Python, PyTorch, vLLM, Unsloth*

Jan. 2026 – April 2026

- Worked on project with other MIT freshmen, whom were all Olympiad winners from many countries. My role was focused most on scaffolding; involved training, hidden state extraction, parameter sweeping, etc.
- Authored our comprehensive technical writeup

Small Learning Model Speech Coach | *Python, PyTorch, Data Gen, Evals*

July 2026

- Distilled a metrics-grounded teacher Gemini 3.1 into Gemma-4-E4B
- Created a harness to generate data by degrading normal speaking audio, measuring speaking metrics, and verifying for good feedback. Also created an eval pipeline for hallucinations in finetuned models.

AWARDS

Putnam Top 200 2026

USAMO Silver 2025

USAPhO Honorable Mention 2025

Two-time Mathematical Olympiad Program (MOP) Attendee 2023, 2024

TECHNICAL SKILLS

Languages: Python, C++, C, RISC-V Assembly, Minispec, Java, Rust, HTML/CSS, JavaScript, TypeScript,

AI Research: Diffusion models, SpecDec, vLLM

Data Engineering: NumPy, SciPy, Pandas, Matplotlib, Jupyter, Automated Eval Pipelines